

Uncertainty-Aware Multi-Spectral Inertial Odometry for Extreme Environments

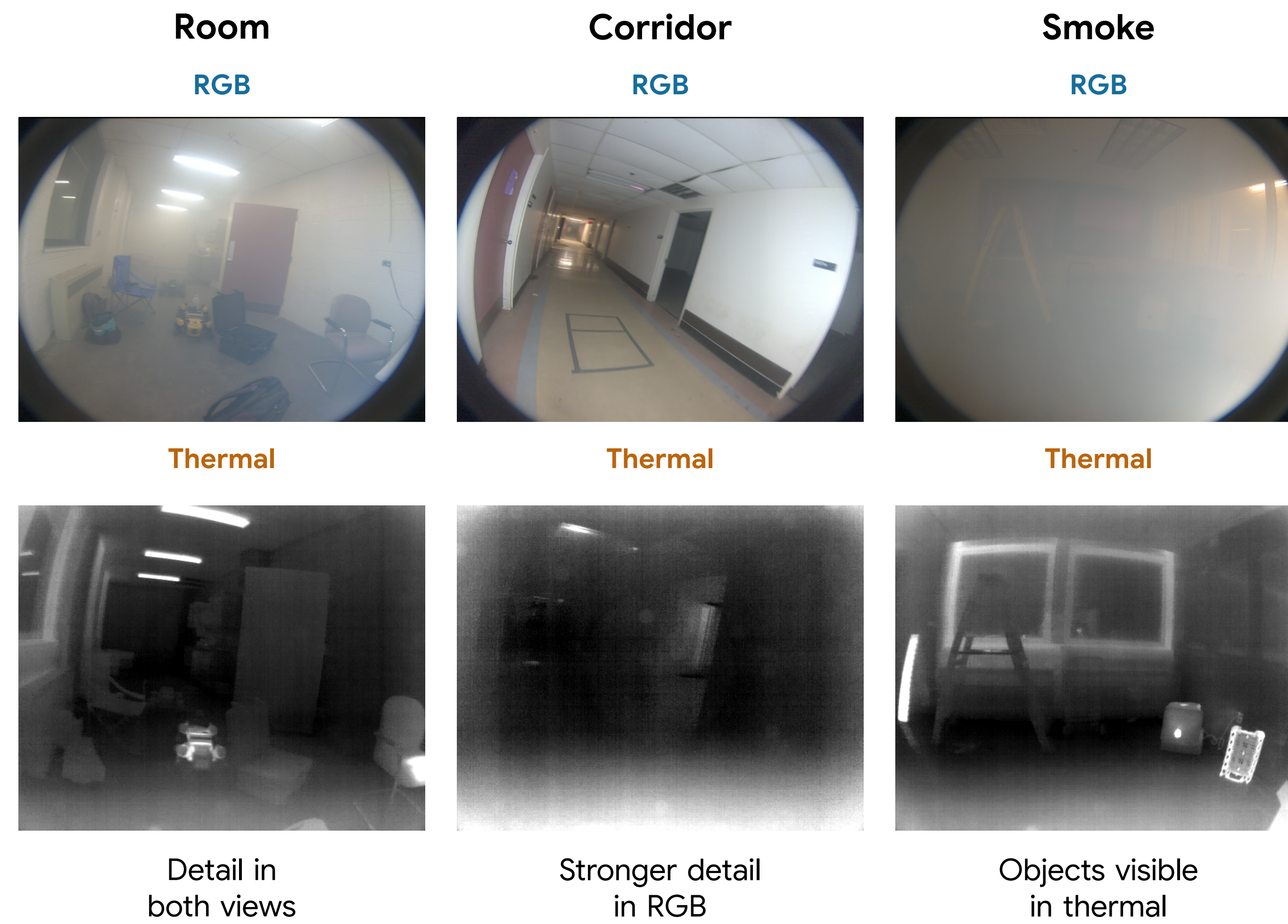
Shibo Zhao, T Karan Balakumar, Avnish Patel, Parv Maheshwari, Tejas Salian, Andrew Jong, Mukai Yu, Wenshan Wang, Sebastian Scherer

1 When Sensor Fusion Hurts

Camera quality changes. Smoke and poor lighting weaken RGB; low thermal contrast weakens thermal. Neither is always better.

More features can reinforce the same motion error. If many features support a biased estimate, adding more does not remove the bias.

Fixed weights can let a degraded camera dominate. Robust losses weight individual residuals; no camera weight is estimated.



SubT-MRS [2]; thermal contrast adjusted for display.

Negative transfer: fusion is less accurate than the best camera-IMU.

4 Accuracy across sequences

Position ATE RMSE [m]; mean of 3 real-time replays.
SE(3) alignment; no loop closure; one Active configuration.
Passive: Huber. Active: DCS + reliability estimation.

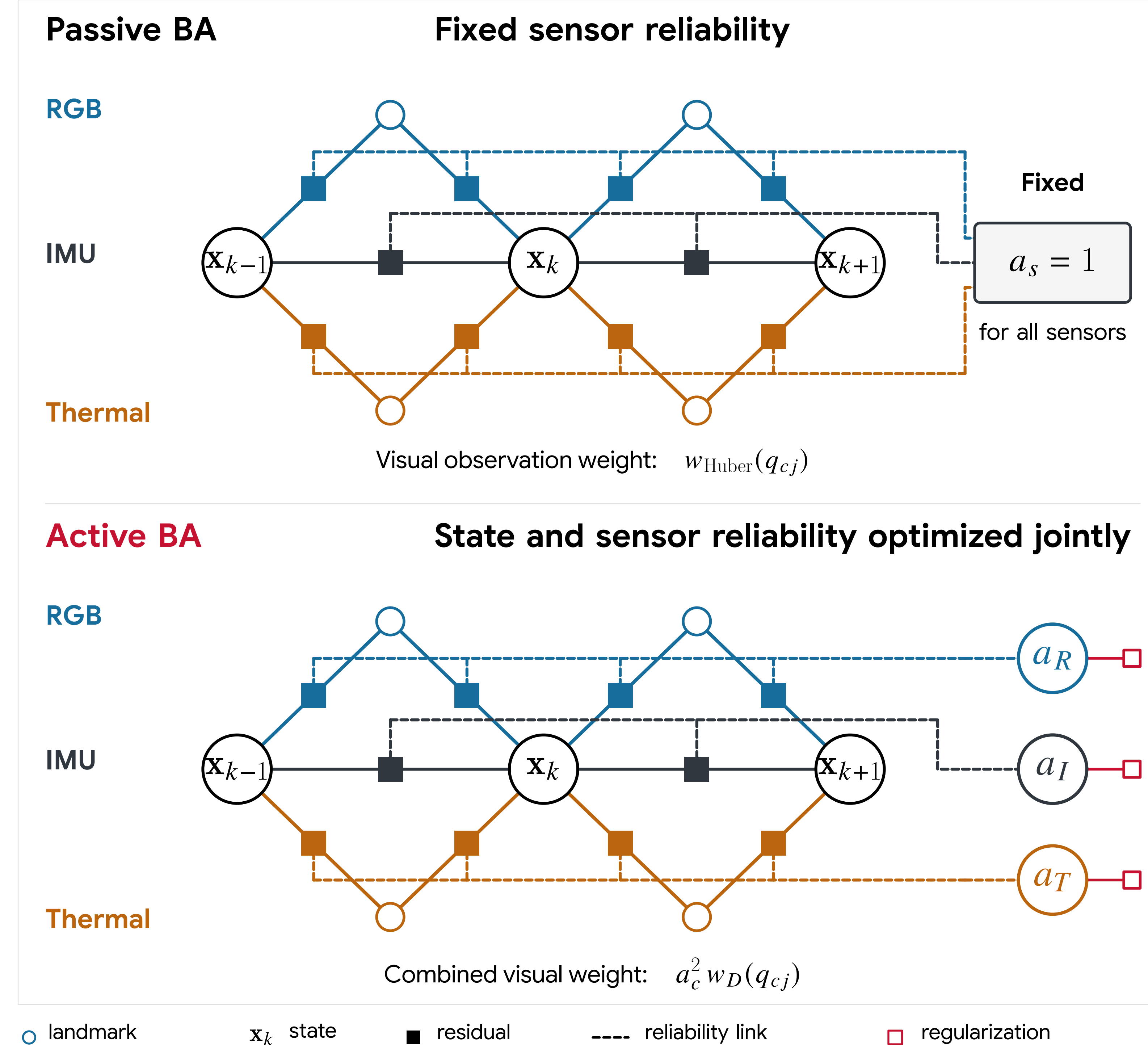
Best single: min(RGB-IMU, thermal-IMU) per sequence; red: fusion > best single.

SubT [2]				CART [3]			
Seq.	Best single	Passive	Active	Seq.	Best single	Passive	Active
10	0.579	0.272	0.217	C1	1.420	1.780	1.380
11	0.238	0.273	0.293	C2	0.368	0.405	0.345
12	0.672	1.008	0.455	C3	0.710	0.568	0.500
13	0.462	0.206	0.208	C4	0.750	0.598	0.587
14	0.157	0.147	0.156	D1	0.295	0.276	0.245
15	0.285	0.347	0.306	D2	0.586	0.612	0.465
Mean	0.399	0.376	0.273	Mean	0.688	0.707	0.587

Mean ATE falls **27% on SubT** and **17% on CART**. 9/12 improve; negative transfer falls from 6/12 to 2/12.

2 From fixed sensor weights to joint reliability estimation

Active BA estimates motion, scene structure and sensor reliability together in the same sliding-window optimization.



Combined regularized objective

$$\min_{X, a} E_0(X) + \frac{1}{2} \sum_{s \in \{R, T, I\}} \left[\underbrace{a_s^2 E_s(X)}_{\text{weighted residual cost}} + \underbrace{\lambda_s (1 - a_s)^2}_{\text{Sensor Reliability Regularization towards 1}} \right]$$

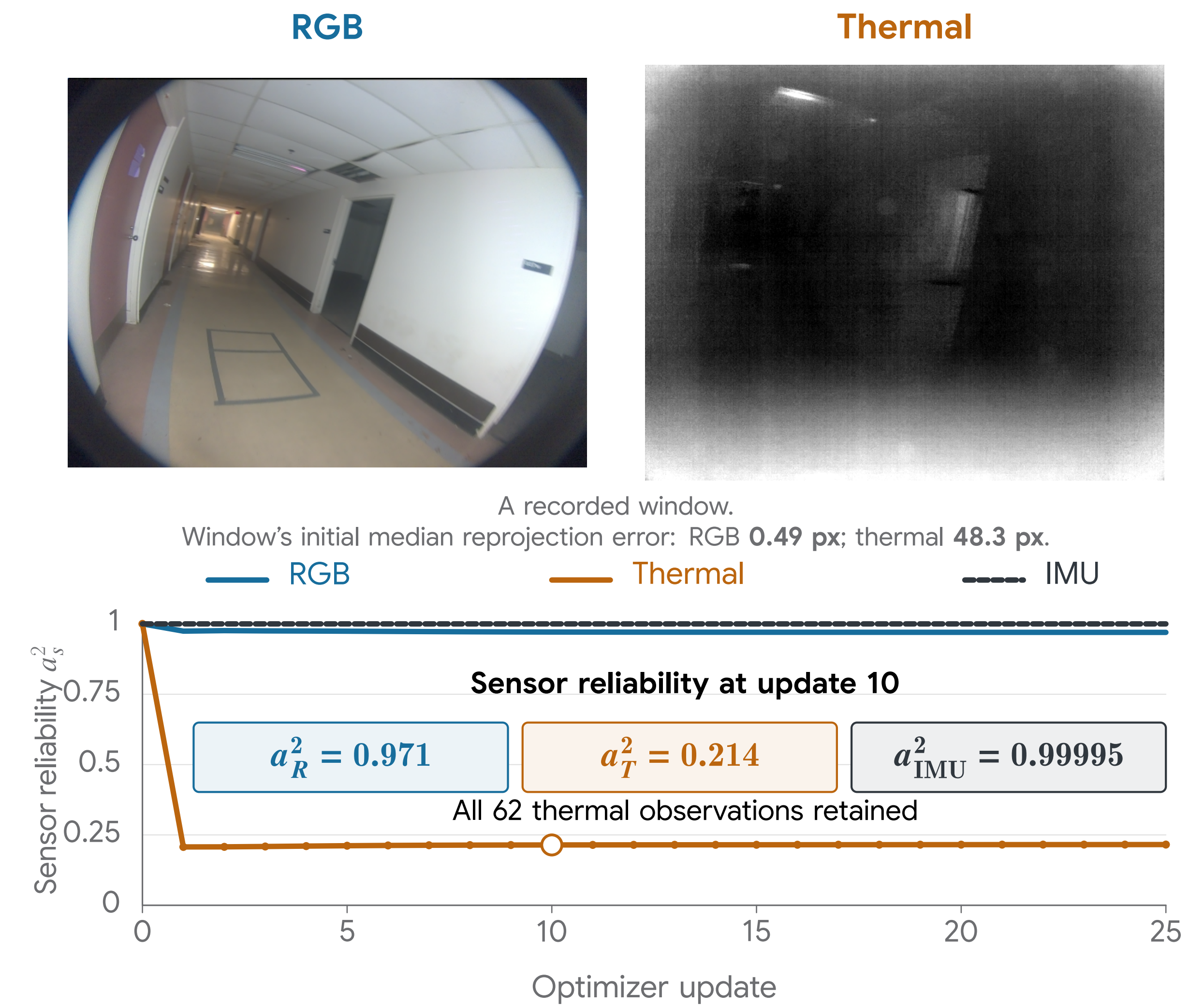
Lowering sensor reliability reduces its residual cost but increases its penalty. The solver balances both while updating the state.

Variables and terms

X : window poses, velocities, IMU biases, inverse depths and enabled calibration.
 $a = (a_R, a_T, a_I)$: sensor reliability parameters; $a_R, a_T \in [0, 1]$, $a_I \in [0.1, 1]$.
 E_s : sum of residual costs: DCS for cameras, quadratic for the IMU.
 E_0 : marginalization prior and auxiliary constraints.
 λ_s : prior strength; $\lambda_c = N_c \eta$ (N_c observations; shared η); λ_I fixed within a solve.
 $s = R, T, I$: RGB, thermal, IMU ($a_{\text{IMU}} = a_I$); c : camera; j : observation.
 q_{cj} : squared, noise-normalized residual; $w_D(q_{cj})$: its DCS weight [1].

3 Inside one solve

Thermal influence falls as its residuals disagree with the state.



5 Wildfire deployment

172.87 m flight; 4.08 m endpoint error (2.36% of distance).

